📄 PlacementConnect - Business Requirements Document (BRD)

1. Executive Summary

PlacementConnect is a web-based campus placement management system intended to coordinate placement activities between an institute placement administrator, recruiting companies and students. It centralises company onboarding, placement-drive publication and approval, student profiles, applications, interview outcomes and placement offers.

The system replaces fragmented coordination through spreadsheets, email and informal communication with role-specific dashboards and controlled workflows. Companies can create and manage drives and process applicants. Students can discover approved opportunities, apply when eligible and within the application window, and view their progress. Institute administrators supervise company and drive approvals, manage user accounts and monitor platform activity.

2. Business Context and Problem Statement

Campus recruitment involves multiple stakeholders and linked activities. Without a shared system, placement teams may need to maintain separate records for company details, student eligibility, job opportunities, applications, interview schedules and final outcomes. This can create duplicated data, inconsistent status reporting, delayed communication and limited visibility.

Problem statement: The institute needs a centralised system to administer campus recruitment while giving companies and students controlled access to the parts of the process relevant to them.

3. Business Objectives

ID	Objective	Intended business outcome
BO-01	Centralise placement information	Reduce fragmented records and improve consistency
BO-02	Control company onboarding	Ensure company accounts are reviewed before participating
BO-03	Govern placement-drive publication	Ensure drives are approved before

		students can discover them
BO-04	Make opportunities discoverable	Allow students to browse approved companies and drives
BO-05	Enforce application eligibility and dates	Reduce invalid applications
BO-06	Track recruitment progress	Maintain application, interview and placement stages
BO-07	Provide role-specific oversight	Give each stakeholder relevant dashboards and operational views
BO-08	Improve operational communication	Provide inbox notifications, reminders and report/export workflows

4. Stakeholders and User Classes

4.1 Institute Administrator

Represents the institute placement administrator. The administrator oversees company registrations, drive approvals, student and company account status, and application information. Admin registration is not exposed in the public registration service; administrator provisioning is therefore outside the documented self-registration flow.

4.2 Recruiting Company

A company user registers, completes its company profile, and subject to institute approval, uses the system to create drives, view applicants and progress application outcomes through interview and placement stages.

4.3 Student

A student user registers, completes a placement profile including academic and resume information, browses approved companies and drives, applies to eligible drives and tracks recruitment outcomes.

5. Scope

5.1 In Scope

- Student and company registration and role-specific login.
- Session-based authentication, logout and current-user retrieval.
- Student and company profile viewing, updating and deletion/export workflows.
- Admin dashboards and management of companies, students, drives and applications.
- Company approval, rejection, reapproval, enable/disable, blacklist/whitelist and deletion.
- Drive creation, update, deletion, approval, rejection and reapproval.
- Student discovery of approved companies and approved drives.
- Drive eligibility checks using minimum CGPA.
- Application submission and withdrawal subject to time and status rules.
- Company review of applications and progression through shortlist, rejection, interview, selection and placement outcomes.
- Role-specific inbox, notification detail and attachment retrieval.
- Dashboard metrics, caching, scheduled/background processing and report/export support.

5.2 Out of Scope

Support for automated resume parsing, AI candidate ranking, video interviews, calendar integrations, offer-letter generation/e-signature, multi-institute tenancy, mobile applications, or a public job-board integration are not implemented.

6. High-Level Business Process

1. A company creates an account and completes its company profile.
2. The institute administrator reviews and approves or rejects the company.
3. An approved company creates a placement drive.
4. The administrator reviews and approves or rejects the drive.
5. Students with active accounts browse approved companies and drives.
6. A student views the drive details and applies only while the application window is open and if the CGPA criterion is met.
7. The company reviews applications and may reject or shortlist an applicant.
8. Shortlisting requires interview mode and interview start/end date-times.
9. The company updates the interview result. A selected candidate requires a joining date and a placement offer record is created.
10. The company updates the offer outcome to Accepted or Declined.
11. The student and administrator can view application-related details through their role-specific views.

7. Business Rules

ID	Rule
BR-01	Public registration is permitted for Student and Company roles only.
BR-02	A user must log in through the role matching the account role.
BR-03	Disabled and blacklisted accounts cannot log in.
BR-04	A company with Pending or Rejected approval cannot log in through the company flow.
BR-05	Students are shown companies only when the company is Approved and its user account is Active.
BR-06	Students are shown drives only when the drive is Approved and its company is Approved and Active.
BR-07	A drive is Upcoming before application_start, Ongoing from application_start through application_deadline, and Closed after the deadline.
BR-08	A student is eligible when minimum_cgpa is not specified or the student's CGPA is greater than or equal to it.
BR-09	An application can be submitted only during the drive's application window.
BR-10	A student cannot submit more than one application for the same drive.
BR-11	A student may withdraw only while the drive is within its application window and the application remains Applied.

BR-12	A company can shortlist an Applied application only when interview mode and interview start/end date-times are supplied.
BR-13	Application status transitions from Applied to Shortlisted or Rejected. Processed application statuses cannot be changed to another status through the current service logic.
BR-14	Interview status begins as Scheduled and may progress to Selected or Not Selected. Selecting requires a joining date.
BR-15	Placement status begins as Offered and may progress to Accepted or Declined. Final statuses cannot be changed to the other final status through the current service logic.
BR-16	A student-drive pair is unique in the database.
BR-17	Dates and date-times are validated against the current IST time in the relevant service flows.

8. Assumptions, Dependencies and Constraints

Assumptions

- One institute placement office operates the instance.
- Admin accounts are provisioned through a controlled process outside public registration.
- The company and student profile forms are completed after account creation.
- The institute determines the policy for approving companies and drives.

Dependencies

- Flask backend.
- Vue frontend.
- Redis for caching and Celery for background task processing.

- Mail integration for notification delivery.
- File storage for generated exports, reports and notification attachments.

Constraints

- The database models use SQLite-compatible configuration; production database is not implemented.
- The application is configured in IST datetime.

9. Risks and Mitigations

Risk	Potential impact	Suggested mitigation
Incomplete student/company profile after registration	Users may be unable to participate meaningfully	Add completion indicators and block relevant actions with clear guidance
Incorrect company or drive approval	Untrusted or invalid opportunities may become visible	Add reviewer guidance, audit trail and approval notifications
Ambiguous status transitions	Stakeholders may misunderstand application progress	Publish a status glossary and show transition history
Date/time boundary confusion	Missed or prematurely closed applications	Display timezone explicitly and test boundary conditions
Account deletion cascading to related records	Loss of recruitment history	Define retention/archive policy and confirm deletion semantics
Notification or background task failure	Missed updates or delayed exports	Add task monitoring, retry policy and delivery status
Sensitive student data exposure	Privacy and reputational risk	Enforce server-side authorisation, least privilege and secure file access

10. Acceptance at Business Level

The solution is considered to meet the current business scope when:

1. Student and company users can register and authenticate through the correct role flows.
2. Administrators can govern company and drive participation.
3. Students can discover approved opportunities and submit valid applications.
4. Companies can manage applicant progress through interview and placement outcomes.
5. Each role can access only its authorised operational views and data.
6. Dashboards provide the operational visibility represented by the application.